OBJECT-CENTRIC SURVEILLANCE INTELLIGENCE (OCSI): A UNIFIED MEMORY-DRIVEN FRAMEWORK FOR ROBUST HUMAN ACTIVITY UNDERSTANDING

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Abstract

Human activity understanding in intelligent surveillance depends on the reliable continuity of object identity across time. Existing pipelines usually optimise object detection, multi-object tracking, person re-identification and activity recognition as loosely coupled stages. This modular design permits detection and association errors to propagate into fragmented trajectories and incomplete behavioural sequences, particularly under prolonged occlusion, crowd congestion, appearance similarity and abrupt motion. This paper introduces Object-Centric Surveillance Intelligence (OCSI), a closed-loop framework that represents each observed person as a persistent semantic entity rather than a sequence of independent frame-level detections. OCSI integrates five interacting layers: perceptual, identity, memory, behaviour and cognitive intelligence. Its central component, the Object Memory Bank, maintains confidence-weighted appearance, motion, trajectory, pose, interaction and behavioural representations for each active or temporarily inactive identity. A behaviour-guided association mechanism then feeds high-confidence activity evidence back into data association to refine identity recovery after occlusion. The framework is formulated through a unified association score, confidence-gated memory updates and a multi-task objective combining detection, identity, memory-consistency, behaviour and temporal losses. The manuscript also defines a reproducible evaluation protocol covering multi-object tracking, re-identification, activity recognition, computational efficiency, statistical significance and failure analysis. Preliminary results reported in the source manuscript indicate improved MOTA, HOTA, IDF1 and activity-recognition accuracy at near-real-time speed; however, these values require independent reproduction on explicitly identified benchmarks before they can support definitive claims. OCSI provides a technically structured route from detection-centric surveillance towards persistent, behaviour-aware object reasoning.

Keywords: object-centric surveillance; multi-object tracking; human activity understanding; person re-identification; object memory bank; behaviour-guided association; temporal memory; intelligent video analytics

1. Introduction

The rapid expansion of camera networks in transportation systems, public spaces, healthcare facilities, industrial sites and critical infrastructure has created a demand for automated systems that can move beyond object localisation to sustained interpretation of human behaviour. Human Activity Understanding (HAU) combines perception, temporal reasoning and contextual interpretation to identify what individuals are doing, how their actions evolve and whether those actions create operational or security risks.

Deep learning has improved the principal components of video surveillance, including object detection, multi-object tracking (MOT), person re-identification (Re-ID) and human activity recognition (HAR). Nevertheless, most operational architectures remain detection-centric. They first detect objects, then associate detections into tracks, and finally apply activity-recognition models to the resulting trajectories. Each stage is usually trained and evaluated independently. Consequently, missed detections, identity switches and fragmented trajectories are inherited by downstream behaviour models, which receive incomplete or incorrectly assigned temporal evidence.

This limitation is especially serious in crowded scenes. A person may be partially visible, disappear behind another individual, change pose or illumination, and reappear after several seconds. Appearance and motion cues alone may be insufficient to recover the correct identity. At the same time, the individual's recent behaviour, interaction partners, direction of travel and activity progression may provide useful semantic evidence. Conventional pipelines rarely use such evidence to correct tracking decisions.

This paper therefore proposes Object-Centric Surveillance Intelligence (OCSI), a unified framework in which every detected person is represented as a persistent semantic entity. The representation is continuously updated with appearance, motion, trajectory, pose, interaction, behaviour and confidence information. OCSI establishes a closed information loop: tracking supports activity recognition, while high-confidence behavioural evidence returns to the association stage to refine identity preservation. This design aims to reduce identity fragmentation and improve the temporal completeness of activity understanding.

The study is guided by the following research question: Can persistent object-level memory and confidence-gated behavioural feedback improve identity preservation and human activity understanding without making surveillance inference computationally impractical?

1.1 Contributions

1. A unified Object-Centric Surveillance Intelligence paradigm that connects perception, identity preservation, temporal memory, behaviour interpretation and cognitive reasoning within a closed-loop architecture.
2. A reproducible Object Memory Bank design based on track-indexed key-value records, confidence-weighted exponential updates, bounded temporal queues, retention rules and memory-decay mechanisms.
3. A confidence-gated behaviour-feedback mechanism that modifies detection-to-track association only when behavioural evidence is sufficiently reliable, thereby limiting error amplification from uncertain HAR predictions.
4. A unified mathematical formulation combining appearance, motion, geometry, memory and behaviour similarities with adaptive weights and a multi-task optimisation objective.
5. A complete experimental protocol for MOT, Re-ID, HAR, ablation, statistical significance, computational efficiency, scalability and failure-case analysis.
6. A transparent distinction between preliminary performance values inherited from the source manuscript and the verified benchmark evidence required for final submission.

1.2 Paper organisation

Section 2 reviews related work and establishes the research gap. Section 3 defines the OCSI architecture. Section 4 formalises object representation, memory update, behaviour feedback and optimisation. Section 5 presents the algorithm and implementation requirements. Section 6 specifies the experimental methodology. Section 7 reports the preliminary results retained from the source manuscript and explains the additional evidence required for defensible claims. Sections 8 and 9 discuss failure modes, ethical considerations, limitations and future work, followed by the conclusion.

2. Related Work and Research Gap

2.1 Deep object detection in surveillance

One-stage detectors from the YOLO family have become common in surveillance because they provide a useful accuracy–latency balance. Transformer-based detectors such as RT-DETR offer global spatial modelling and end-to-end set prediction. Despite these advances, object detectors optimise frame-level localisation and do not normally maintain persistent semantic identities. Small targets, partial visibility, poor lighting and dense overlap therefore remain important sources of downstream tracking failure (Jocher et al., 2023; Lv et al., 2023).

2.2 Multi-object tracking

Tracking-by-detection methods associate frame-level detections using motion, geometry and appearance. DeepSORT incorporates deep appearance embeddings; ByteTrack uses both high- and low-confidence detections; BoT-SORT combines appearance, camera-motion compensation and refined motion models; and OC-SORT emphasises observation-centric motion estimation (Wojke et al., 2017; Zhang et al., 2022; Aharon et al., 2022; Cao et al., 2023). These approaches are effective but usually retain short-term descriptors and treat behaviour as a downstream output rather than an association cue.

2.3 Transformer-based tracking

TrackFormer and MOTR reformulate tracking through persistent transformer queries and joint detection-association learning (Meinhardt et al., 2022; Zeng et al., 2022). Their end-to-end formulation reduces reliance on handcrafted association rules, but long-term occlusion, query drift, computational cost and limited integration with explicit human-activity reasoning remain open challenges. OCSI is complementary rather than mutually exclusive: its structured memory and behaviour-feedback principles can be attached to either tracking-by-detection or transformer-based trackers.

2.4 Person re-identification

Re-ID models learn discriminative identity features across time or cameras. Transformer-based representations such as TransReID improve global and local feature modelling, while self-supervised methods reduce dependence on expensive identity labels (He et al., 2021; Yang et al., 2024). Appearance-only Re-ID, however, remains vulnerable when multiple people share similar clothing, body shape or viewpoint. OCSI therefore treats appearance as one component of a larger object memory.

2.5 Human activity recognition and interaction modelling

Modern HAR uses convolutional, recurrent, graph-based and transformer architectures to model spatial and temporal patterns. Although performance has improved, most HAR systems assume that the input person tube or track is correct and temporally continuous. Identity switches can merge actions from different individuals, while fragmented tracks truncate long activities. Existing HAR therefore inherits tracking errors instead of actively correcting them (Bukht et al., 2024; Kaur et al., 2024; Arrotta et al., 2025).

2.6 Object-centric learning and temporal memory

Object-centric learning represents scenes through persistent entities and their relations rather than undifferentiated frame features. Slot-based and memory-driven approaches have shown the value of entity persistence in sequential reasoning (Locatello et al., 2020). Surveillance tracking uses related ideas, but usually in restricted form, such as a recent appearance gallery or trajectory buffer. A persistent multimodal memory combining identity and behaviour remains insufficiently explored.

2.7 Research gap

The literature reveals three connected gaps. First, detection, tracking, Re-ID and HAR are frequently optimised as separate tasks. Second, trackers rarely maintain an explicit, bounded and confidence-aware memory that integrates appearance, trajectory, pose, interactions and behaviour. Third, behavioural evidence is rarely fed back into identity association. OCSI addresses these gaps through persistent object records and confidence-gated closed-loop refinement.

Approach	Detection	Tracking	Re-ID	Persistent memory	HAR	Behaviour feedback	Primary limitation
DeepSORT	External	Yes	Yes	Short-term gallery	No	No	Appearance and linear-motion dependence
ByteTrack	External	Yes	No	No	No	No	No explicit identity memory
BoT-SORT	External	Yes	Yes	Limited	No	No	No behaviour-aware closed loop
OC-SORT	External	Yes	Optional	Trajectory focused	No	No	Limited semantic reasoning
TrackFormer/MOTR	Joint	Yes	Query-based	Latent query state	No	No	Computational cost and limited HAR integration
Proposed OCSI	Yes	Yes	Yes	Explicit multimodal object memory	Integrated	Yes	Requires comprehensive benchmark verification

Table 1. Positioning of OCSI relative to representative surveillance approaches.

3. Proposed Object-Centric Surveillance Intelligence Framework

3.1 Definition and design principles

OCSI is a unified surveillance framework in which every detected person is represented as a persistent semantic entity whose visual appearance, spatial state, motion dynamics, trajectory, pose, contextual relations, behavioural evolution and confidence are jointly maintained to support continuous perception, robust identity preservation and reliable human activity understanding.

- Persistence: an identity record survives temporary observation loss for a controlled retention period.
- Multimodality: identity evidence includes appearance, motion, geometry, pose, context and behaviour.
- Confidence awareness: uncertain observations have reduced influence on memory and association.
- Bounded memory: queues and feature galleries have explicit capacity and decay rules.
- Closed-loop reasoning: behaviour can refine tracking, but only through confidence-gated feedback.
- Modularity: the memory and feedback modules can be attached to conventional or transformer-based trackers.

3.2 Five-layer architecture

Perceptual Intelligence: Performs frame acquisition, preprocessing, object detection, pose estimation and feature extraction. An occlusion-aware multi-scale attention module may be attached to the detector to strengthen partially visible and small-person representations.

Identity Intelligence: Generates preliminary tracks using motion prediction, appearance similarity, geometric overlap and candidate gating. It also resolves identity conflicts and manages active, lost and reactivated tracks.

Memory Intelligence: Maintains one bounded object record per confirmed identity. The layer retrieves relevant historical representations, computes memory compatibility, updates reliable records and decays stale evidence.

Behaviour Intelligence: Recognises actions and interactions from temporally aligned trajectory, pose, visual and relational cues. It produces both class probabilities and a compact behaviour embedding.

Cognitive Intelligence: Aggregates object and scene information for anomaly assessment, risk interpretation and decision support. It also controls whether behaviour feedback is sufficiently reliable to influence tracking.

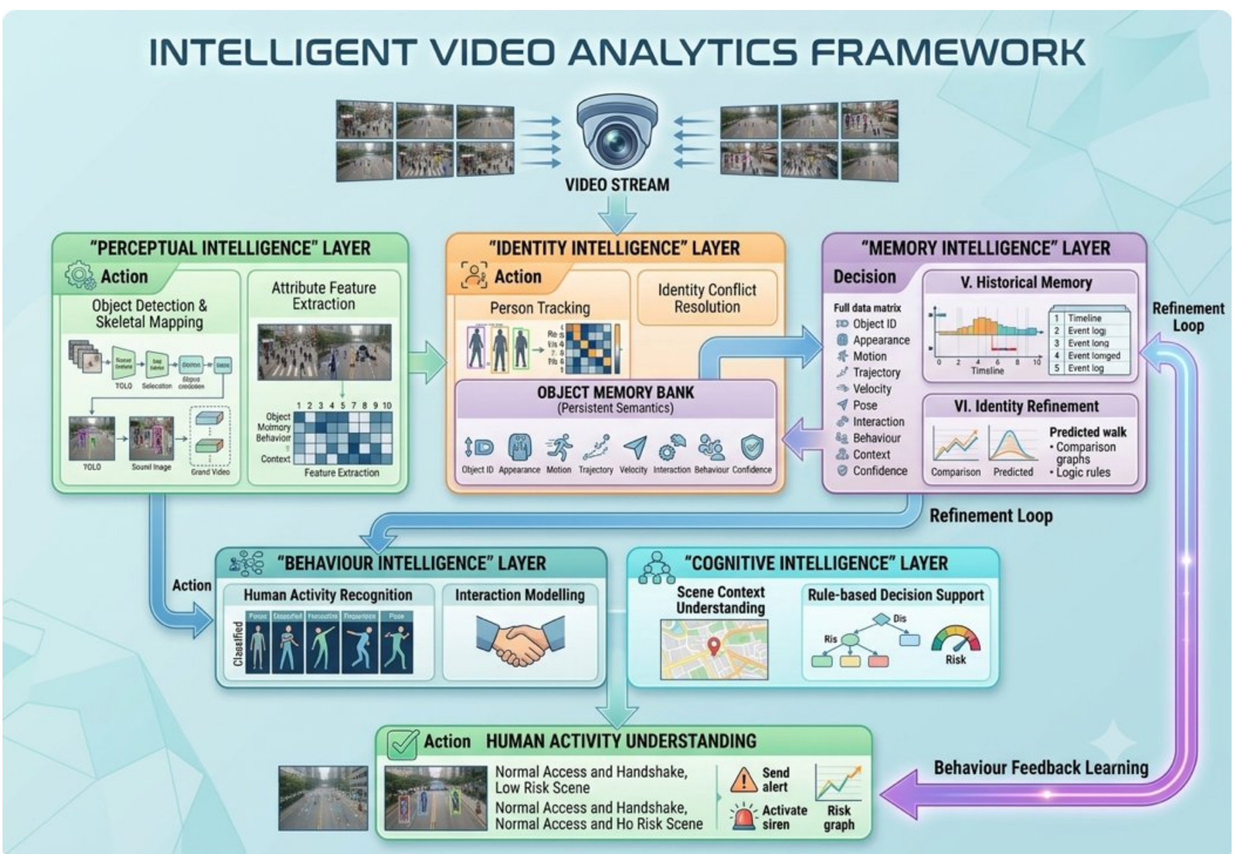

Figure 1. High-level OCSI architecture linking perception, identity, memory, behaviour and cognitive reasoning.

3.3 Object Memory Bank

The Object Memory Bank is implemented as a track-indexed key-value store. Each confirmed identity is a key, and its value is a structured record containing fixed-dimensional vectors and bounded temporal queues. This definition removes ambiguity concerning whether the memory is a conceptual repository or an executable data structure.

$$M_i^t = \{\bar{a}_i^t, G_i^t, Q_{i,\tau}^t, Q_{i,v}^t, Q_{i,p}^t, Q_{i,c}^t, \bar{b}_i^t, q_i^t, s_i^t\}$$

Here, $\bar{a}_i^t$ is the exponentially smoothed appearance prototype; G_i^t is a bounded gallery of recent reliable appearance embeddings; $Q\tau$, Qv, Qp and Qc are queues for trajectory, velocity, pose and contextual relations; $\bar{b}_i^t$ is the behaviour prototype; q_i^t is memory confidence; and s_i^t is the track state, such as tentative, confirmed, lost or archived.

Memory field	Representation	Update rule	Capacity	Purpose
Appearance prototype	d-dimensional vector	Confidence-weighted EMA	1	Long-term identity descriptor
Appearance gallery	Feature queue	Append reliable observations	K_a	Handles viewpoint variation
Trajectory	Bounding-box centres/boxes	FIFO queue	$L\tau$	Motion continuity
Velocity	2D/4D state	Kalman/finite difference	Lv	Dynamic compatibility
Pose	Keypoints or pose embedding	FIFO queue	Lp	Body configuration
Context	Neighbour/interaction embedding	FIFO queue	Lc	Group and scene relations
Behaviour	Probability/embedding	Confidence-weighted EMA	1 + Kb	Semantic continuity
Confidence/state	Scalar and status	Rule-based	1	Controls update and retention

Table 2. Executable structure of the Object Memory Bank.

3.4 Memory creation, update, decay and deletion

7. Creation: initialise a tentative record after an unmatched detection persists for the minimum confirmation period.
8. Update: update only when the association cost is below a reliability threshold and the detection confidence exceeds the update threshold.
9. Appearance update: combine the previous prototype with the new embedding using confidence-weighted exponential smoothing.
10. Queue update: append trajectory, velocity, pose and context observations and remove the oldest element when capacity is exceeded.
11. Lost-state retention: preserve unmatched confirmed tracks for a maximum age so that they can be reactivated after occlusion.
12. Decay: reduce memory confidence as the time since last reliable observation increases.
13. Deletion or archiving: remove records after the maximum retention period or archive them for cross-camera retrieval when multi-camera mode is enabled.
14. Contamination control: freeze or roll back memory updates when a later identity-conflict test indicates a likely incorrect match.

$$\bar{a}_i^t = \alpha_i^t\, \bar{a}_i^{t-1} + (1 - \alpha_i^t)\, \hat{a}_j^{\,t}, \quad \alpha_i^t = clip(\alpha_0 + \kappa(1 - r_{ij}^{\,t}), \alpha min, \alpha max)$$

The reliability score r_{ij}^t combines detection confidence, assignment margin and consistency across motion, appearance and behaviour. Reliable observations receive greater update weight, while ambiguous matches preserve more of the previous memory.

3.5 Behaviour-guided identity refinement

The Behaviour Intelligence layer produces a class-probability vector and a behaviour embedding for each track window. Behaviour does not replace appearance or motion. Instead, it acts as a gated semantic cue. When activity confidence is low, its contribution approaches zero. When confidence is high and the activity sequence is temporally plausible, it can favour or reject candidate associations.

$$S_{ij}^{\,t} = w_a Sapp + w_m Smotion + wgSIoU + whSmemory + wb\cdot g_{ij}^{\,t}\cdot Sbehaviour$$

$$g_{ij}^{\,t} = \mathbb{1}[pmax \geq \theta b] \cdot pmax \cdot exp(-\Delta t/\tau b)$$

The gate g_{ij}^{t} depends on the maximum activity probability, an activity-confidence threshold and the time since the last reliable behaviour observation. This prevents uncertain HAR outputs from corrupting identity association. Behaviour can modify tracking through association refinement, contradiction rejection and reactivation of temporarily lost tracks.

Figure 2. Operational workflow showing memory retrieval, behaviour recognition and feedback-assisted identity association.

4. Mathematical Formulation

4.1 Problem definition

Given a video sequence $V = \{I_1, \ldots, I\_T\}$, the detector returns $D^t = \{d_j^t\}$ at frame t. Each detection contains a bounding box, class confidence and visual embedding. The tracking problem is to assign detections to persistent identities while estimating their activities over time. OCSI seeks assignments that are locally plausible and globally consistent with object memory and behaviour.

4.2 Similarity terms and gating

$$Sapp = \cos(\hat{a}_j^{t}, \bar{a}_i^{t}), \quad Smotion = \exp(-½(z_j-\mu_i)^{\top}\Sigma_i^{-1}(z_j-\mu_i)), \quad SIoU = IoU(B_j^{t}, \hat{B}_i^{t})$$

$$Smemory = \beta_1 \cos(\hat{a}_j^{t}, \bar{a}_i^{t}) + \beta_2 Strajectory + \beta_3 Spose + \beta_4 Scontext$$

$$Sbehaviour = \cos(\hat{b}_j^{t}, \bar{b}_i^{t})$$

Candidate pairs that violate the Kalman gating threshold, class consistency or a minimum geometric plausibility rule are removed before assignment. The remaining similarity matrix is converted to cost and solved using the Hungarian algorithm or a differentiable matching alternative.

4.3 Adaptive weighting

$$w_k^t = softmax(u_k^t), \ \ k \in \{a,m,g,h,b\}$$

The weights may be fixed through validation or predicted by a lightweight reliability network using occlusion ratio, detection confidence, track age, motion uncertainty and behaviour confidence. Under severe occlusion, memory may receive greater weight; under abrupt motion, appearance and pose may dominate; and under uncertain activity prediction, behaviour is suppressed.

4.4 Multi-task objective

$$L = \lambda detLdet + \lambda trkLtrk + \lambda idLid + \lambda memLmem + \lambda behLbeh + \lambda tempLtemp + \lambda gateLgate$$

- Ldet: detector classification, localisation and objectness losses.
- Ltrk: association or assignment loss.
- Lid: metric-learning or classification loss for identity discrimination.
- Lmem: consistency between current reliable observations and stored identity prototypes.
- Lbeh: activity classification and interaction-recognition loss.
- Ltemp: temporal smoothness or sequence-consistency loss.
- Lgate: calibration loss ensuring that feedback confidence reflects prediction correctness.

4.5 Complexity

Let N be the number of active tracks, M the number of detections, d the embedding dimension and L the temporal window. Detection remains the dominant convolutional or transformer cost. After gating, association has worst-case cubic complexity in max(N, M), although practical matrices are sparse. Memory retrieval and update are O(Nd), while fixed-window behaviour inference is approximately O(NLdb) for a model-dependent constant db. The added memory overhead is therefore linear in the number of active tracks when capacities are fixed.

5. Algorithm and Implementation

5.1 Inference algorithm

1. Initialise the detector, pose or feature extractor, tracker, behaviour network and empty Object Memory Bank.
2. For each frame, preprocess the image and detect persons.
3. Extract appearance and optional pose embeddings for every detection.
4. Predict active-track locations and uncertainties using the motion model.
5. Retrieve object-memory records for active and retained lost tracks.
6. Apply motion, class and geometry gating to construct candidate detection–track pairs.
7. Calculate appearance, motion, IoU, memory and confidence-gated behaviour similarities.
8. Form the adaptive association cost matrix and solve the assignment problem.
9. Update confirmed identities only when assignment reliability exceeds the memory-update threshold.
10. Move unmatched confirmed tracks to the lost state; retain them until the maximum age is reached.
11. Initialise tentative tracks from unmatched detections and confirm them after the minimum hit count.
12. Run activity recognition over each sufficiently long track window.
13. Update the behaviour prototype when activity confidence exceeds the behaviour threshold.
14. Re-evaluate ambiguous or reactivation matches with behaviour evidence.
15. Decay stale memory, archive or delete expired records, and output tracks and activity labels.

5.2 Training strategy

A practical implementation can use staged training followed by joint fine-tuning. First, train or initialise the detector on person-detection data. Second, train the Re-ID encoder using supervised or self-supervised metric learning. Third, train the behaviour network on temporally aligned person tracks. Fourth, initialise memory and feedback components using frozen backbones. Finally, jointly fine-tune the differentiable modules with the unified objective while retaining non-differentiable Hungarian assignment at inference.

5.3 Required reproducibility information

Mandatory before submission: The following information is absent from the source manuscript and must be completed from the actual implementation. Do not replace it with estimates.

Item	Value to insert	Verification source
Code repository and commit hash	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Python, CUDA and deep-learning framework versions	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Detector and Re-ID backbone versions	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Behaviour-network architecture and input window	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Input resolution and frame sampling rate	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Training datasets and exact train/validation/test splits	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Pretraining sources and checkpoint selection	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Optimiser, learning rate, scheduler, epochs and batch size	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Data augmentation	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Association weights and gating thresholds	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Memory capacities, EMA coefficient and retention age	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Random seeds and number of independent runs	[AUTHOR TO COMPLETE]	Configuration file, log or repository
GPU, CPU, RAM and operating system	[AUTHOR TO COMPLETE]	Configuration file, log or repository
Official evaluation toolkit versions	[AUTHOR TO COMPLETE]	Configuration file, log or repository

Table 3. Reproducibility information required before submission.

6. Experimental Methodology

6.1 Benchmark strategy

A defensible evaluation must separate component-level performance from end-to-end surveillance performance. MOT benchmarks test identity continuity, Re-ID datasets test appearance recovery, HAR datasets test action recognition, and unified videos with both identity and activity labels test the central closed-loop claim. The paper should not imply that a result obtained on one dataset validates all components.

Task	Recommended benchmark	Purpose	Primary metrics	Status in source draft
MOT	MOT17, MOT20, DanceTrack	General, crowded and appearance-ambiguous tracking	HOTA, DetA, AssA, MOTA, IDF1, IDs, FPS	Not identified
Detection	CrowdHuman or MOT17Det	Crowded-person detection	mAP, precision, recall	Not identified
Re-ID	Market-1501, MSMT17, MARS	Cross-view and video identity retrieval	Rank-1, mAP	Not identified
HAR	UCF101, HMDB51, NTU RGB+D or AVA	Activity classification/detection	Accuracy, macro-F1, mAP	Not identified
Unified OCSI	Dataset containing person IDs and activity annotations	Tests behaviour-to-tracking feedback	IDF1, IDs, HAR F1, joint score	Required

Table 4. Recommended benchmark coverage and validation purpose.

6.2 Baselines

The tracking comparison should include conventional and transformer-based baselines under identical test conditions. At minimum, evaluate DeepSORT, ByteTrack, BoT-SORT and OC-SORT, together with TrackFormer and MOTR or more recent publicly reproducible query-based trackers. The primary OCSI comparison should use the same detector wherever possible so that gains can be attributed to memory and behaviour rather than detector differences.

6.3 Evaluation metrics

- Tracking: HOTA, DetA, AssA, MOTA, IDF1, identity switches, fragmentations and mostly tracked/lost trajectories.
- Detection: mAP at standard IoU thresholds, precision, recall and F1-score.
- Re-ID: cumulative matching characteristic Rank-1/Rank-5 and mean average precision.
- HAR: accuracy, macro-precision, macro-recall, macro-F1 and per-class confusion matrix.
- Efficiency: end-to-end latency, FPS, parameters, FLOPs, GPU memory and energy where available.
- Recovery: identity recovery rate stratified by occlusion duration.
- Calibration: expected calibration error for behaviour confidence and feedback gating.

6.4 Ablation and sensitivity analysis

15. Baseline detector plus tracker.
16. Add occlusion-aware feature refinement.
17. Add appearance-only memory.
18. Add trajectory, pose and context memory.
19. Add ungated behaviour feedback.
20. Add confidence-gated behaviour feedback.
21. Vary appearance-gallery capacity, temporal-window length, EMA coefficient and maximum track age.
22. Vary behaviour threshold and adaptive-weight strategy.
23. Evaluate under synthetic or annotated occlusion levels and crowd densities.
24. Replace the conventional tracker with a transformer tracker to test architectural portability.

6.5 Statistical analysis

Each trainable configuration should be executed with at least five independent random seeds when computationally feasible. Report mean, standard deviation and 95% confidence intervals. Use paired comparisons over identical sequences. A paired t-test is suitable when difference scores are approximately normal; otherwise use the Wilcoxon signed-rank test. Report exact p-values and an effect size, such as Cohen's d or rank-biserial correlation. For sequence-level MOT metrics, bootstrap confidence intervals across videos provide an additional robust analysis.

6.6 Real-time evaluation

Real-time claims must be based on end-to-end inference, not detector-only speed. Report hardware, input resolution, batch size, numerical precision, warm-up period, number of timed frames and whether video decoding is included. At 30 FPS, average end-to-end latency should be approximately 33.3 ms per frame or lower. Scalability should be tested as the number of visible people increases.

7. Results and Discussion

Status of the following tables: These values are retained from the source manuscript as preliminary reported results. They must be linked to named datasets, official evaluation scripts, experiment logs and repeated runs before submission. Until then, the text must avoid definitive claims of state-of-the-art superiority.

7.1 Preliminary tracking results

Method	MOTA (%)	HOTA (%)	IDF1 (%)	IDs ↓	Precision (%)	Recall (%)	FPS
DeepSORT	77.8	61.5	75.3	684	89.2	83.8	35
ByteTrack	80.6	64.7	79.5	492	91.4	86.5	30
BoT-SORT	81.7	66.8	81.4	418	92.1	87.2	28
OC-SORT	82.3	67.9	82.6	386	92.8	87.9	31
Proposed OCSI	85.4	72.1	87.8	214	94.3	90.5	29

Table 5. Preliminary tracking results reported in the source manuscript.

The preliminary table suggests that OCSI improves identity-oriented metrics more strongly than detection-oriented metrics. Relative to OC-SORT, the reported IDF1 gain is 5.2 percentage points and identity switches decrease from 386 to 214. This pattern is consistent with the intended role of persistent memory and feedback. Nevertheless, the claim remains provisional because the dataset, detector parity, evaluation toolkit and variance across runs are not supplied.

7.2 Preliminary component results

Task/configuration	Metric 1	Metric 2	Metric 3	Metric 4
Detection: YOLOv8	mAP@0.5 91.6	Precision 91.8	Recall 88.7	F1 90.2
Detection: YOLO11	mAP@0.5 92.4	Precision 92.5	Recall 89.4	F1 90.9
Detection: OCSI + OMA-Net	mAP@0.5 94.1	Precision 94.3	Recall 91.8	F1 93.0
Re-ID: OCSI	Rank-1 91.8	mAP 85.7	ID Recall 90.6	—
HAR: OCSI	Accuracy 96.3	Precision 95.8	Recall 95.1	F1 95.4

Table 6. Preliminary detection, Re-ID and HAR values retained from the source manuscript.

These component results cannot be interpreted rigorously until each row is connected to a specific dataset and protocol. In particular, Rank-1 and mAP values depend strongly on the Re-ID benchmark, while HAR accuracy depends on class count, split and whether evaluation is person-level, clip-level or frame-level.

7.3 Preliminary ablation study

Configuration	MOTA (%)	IDF1 (%)	IDs ↓
YOLO11 + DeepSORT	80.8	80.2	452
+ OMA-Net	82.3	81.8	398
+ Object Memory Bank	84.1	85.1	271
+ Behaviour Feedback	84.8	86.6	233
Full OCSI	85.4	87.8	214

Table 7. Preliminary ablation results reported in the source manuscript.

The reported ablation pattern attributes the largest reduction in identity switches to the Object Memory Bank, followed by a further gain from behaviour feedback. A stronger final paper should separate appearance memory from trajectory, pose, context and behaviour memory; compare gated and ungated feedback; and report mean ± standard deviation across repeated runs.

7.4 Transformer comparison to be added

The source manuscript discusses TrackFormer and MOTR but does not include them experimentally. This omission prevents a fair assessment against modern end-to-end tracking. The final comparison should use the same dataset, resolution, detector policy and hardware and should report both accuracy and efficiency.

Method	Backbone/detector	HOTA	AssA	IDF1	IDs	FPS	Parameters
TrackFormer	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]
MOTR	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]
Recent memory/query tracker	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]
OCSI with conventional tracker	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]
OCSI with transformer tracker	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]

Table 8. Required controlled comparison with transformer-based tracking methods.

8. Failure Analysis, Limitations and Ethical Considerations

8.1 Expected and observed failure categories

A submission-ready study must disclose where the method fails. OCSI may be vulnerable to the following conditions, which should be tested with sequence examples and stratified metrics.

Failure condition	Likely mechanism	Diagnostic measure	Mitigation
Incorrect HAR prediction	Wrong behaviour cue biases association	Feedback error rate and ID switches after HAR error	Confidence gate and rollback

Memory contamination	Incorrect match updates identity prototype	Prototype drift and assignment margin	Reliable-update threshold and dual gallery
Very long occlusion	Stored evidence becomes stale	Recovery rate by occlusion length	Decay-aware reactivation and archive search
Similar appearance and behaviour	All cues become ambiguous	Pairwise confusion cases	Context, pose and longer temporal evidence
Abrupt activity change	Behaviour history becomes misleading	Transition-specific errors	Fast behaviour adaptation and lower gate weight
Dense crowd overlap	Detection and pose fail simultaneously	Performance by crowd density	Stronger detector and uncertainty-aware association
Cross-camera domain shift	Appearance and geometry change	Camera-pair Re-ID performance	Domain adaptation and camera-aware memory
Rare or unseen activities	Behaviour network is poorly calibrated	Per-class recall and calibration error	Open-set recognition and uncertainty suppression

Table 9. Failure modes, diagnostics and proposed mitigation strategies.

8.2 Required qualitative analysis

The final paper should include frame sequences comparing a baseline tracker with OCSI under short occlusion, long occlusion, group crossing, similar clothing, abrupt direction change and incorrect behaviour prediction. Each example should display track IDs, activity confidence, memory confidence and the association decision. Both successful recovery and failure cases should be shown.

8.3 Limitations

- The current framework is modular rather than fully end-to-end, and some assignment operations are non-differentiable.
- Behaviour feedback can amplify errors when the HAR model is miscalibrated, although confidence gating is designed to reduce this risk.
- Memory increases storage and state-management complexity as crowd size and retention duration grow.
- Cross-camera persistence requires camera-aware normalisation and privacy-sensitive identity management beyond the present single-stream formulation.
- The preliminary numerical results in the source draft lack the dataset and reproducibility information required for independent verification.
- The framework has not yet been compared experimentally with recent transformer-based trackers under controlled conditions.

8.4 Ethical, privacy and deployment considerations

Persistent object memory raises privacy and governance concerns because identity-related representations may be retained beyond the immediate frame. Deployment should therefore apply data minimisation, short retention periods, encryption, access controls, audit logs and clear legal authority. Where possible, memory should store non-invertible embeddings rather than raw biometric imagery. The system should be evaluated for performance disparities across demographic and environmental groups, and high-stakes alerts should remain subject to human review. OCSI is intended to support situational understanding, not autonomous punitive decision-making.

9. Conclusion and Future Work

This paper introduced Object-Centric Surveillance Intelligence, a closed-loop framework that represents each observed person through persistent multimodal memory and uses confidence-gated behavioural evidence to refine identity association. The revised formulation specifies the Object Memory Bank as an executable track-indexed data structure with bounded feature galleries, temporal queues, reliability-aware updates, decay and retention rules. It also clarifies how behaviour modifies tracking through gated similarity, contradiction rejection and lost-track reactivation. These elements provide a stronger technical foundation than a purely conceptual architecture.

The preliminary results inherited from the source manuscript suggest potentially meaningful gains in identity preservation and activity recognition while maintaining near-real-time throughput. Nevertheless, the absence of named datasets, implementation settings, repeated runs and transformer baselines prevents definitive claims. The immediate priority is therefore empirical completion: reproduce the system on identified MOT, Re-ID and HAR benchmarks; implement a unified identity-and-activity evaluation; report statistical significance; analyse failures; and release code and configurations.

Future work should investigate a transformer-based OCSI variant with persistent object queries, graph-structured interaction memory, multimodal RGB–thermal sensing, open-set behaviour recognition, privacy-preserving federated learning and resource-aware deployment on edge devices.

Appendix A. Final Submission Completion Checklist

Requirement	Status	Evidence/location
Dataset names and versions stated	☐	
Train/validation/test splits stated	☐	
Official evaluation toolkits stated	☐	
Hardware and software environment stated	☐	
Hyperparameters and thresholds stated	☐	
Random seeds and repeated runs completed	☐	
Mean, standard deviation and confidence intervals reported	☐	
Statistical tests and effect sizes reported	☐	
Transformer baselines included	☐	
Detector parity controlled	☐	
Failure examples included	☐	
Occlusion-stratified analysis included	☐	
Code and model weights archived	☐	
Ethics/privacy statement included	☐	
Duplicate and incomplete references corrected	☐	
All figures regenerated at publication resolution	☐	

Table A1. Completion checklist to be satisfied before journal submission.

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